Stoke's theorem

Stoke's theorem transforms a line integral to a surface integral.

Statement:-

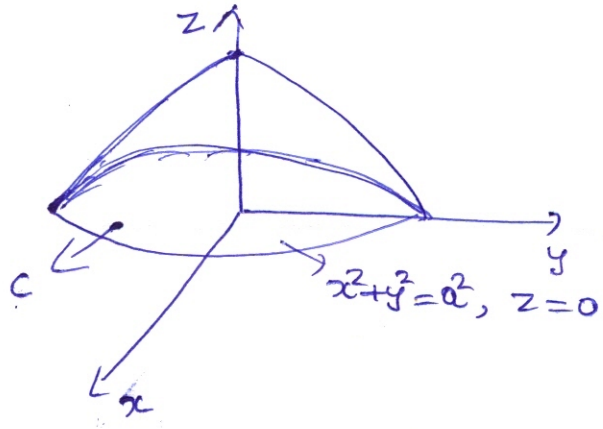
If $\vec{F}$ is any continuously differentiable vector function and S is a surface bounded by a curve C, then

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S \text{Curl}\,\vec{F}\cdot\hat{n}\,ds = \iint_S (\nabla\times\vec{F})\cdot\hat{n}\,ds$$

where $\hat{n}$ is the unit normal at any point S drawn outwardly.

(1) Verify Stoke's theorem for the function $\vec{F} = y^2\hat{i} + xy\hat{j} - xz\hat{k}$ where S is the hemisphere $x^2+y^2+z^2=a^2$ and C is the boundary.

Soln:-



By Stoke's theorem, we have $\oint_C \vec{F}\cdot d\vec{r} = \iint_S \text{Curl}\,\vec{F}\cdot\hat{n}\,ds$ → (1)

The enclosed curve C for the hemisphere is the base circle whose eqn is $x^2+y^2=a^2$

For any point on C, we have

$$x=a\cos\theta, \quad y=a\sin\theta, \quad z=0, \quad 0\le\theta\le 2\pi$$

$$dx=-a\sin\theta\,d\theta, \quad dy=a\cos\theta\,d\theta, \quad dz=0$$

Thus $\oint_C \vec{F}\cdot d\vec{r} = \oint_C (y^2\hat{i}+xy\hat{j}-xz\hat{k})\cdot(dx\hat{i}+dy\hat{j}+dz\hat{k})$

$$\oint_C \vec{F}\cdot d\vec{r} = \oint_C y^2dx+xydy-xzdz$$

$$= \int_{\theta=0}^{\theta=2\pi} a^2\sin^2\theta\,(-a\sin\theta)\,d\theta + (a^2\cos\theta\sin\theta)(a\cos\theta)\,d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi} a^3(-\sin^3\theta+\cos^2\theta\sin\theta)\,d\theta$$

$$= a^3\int_{\theta=0}^{2\pi}(\cos^2\theta\sin\theta-\sin^3\theta)\,d\theta$$

$$\oint_C \vec{F}\cdot d\vec{r} = 0 \quad \rightarrow (2)$$

$\left[\because \int_0^{2a} f(x)\,dx = 0 \right.$ if $f(2a-x)=-f(x$

Now consider

$$\text{curl}\,\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & xy & -xz \end{vmatrix}$$

$= z\hat{j} - y\hat{k}$

let $\phi(x,y,z) = x^2+y^2+z^2-a^2$

$\nabla\phi = 2x\hat{i}+2y\hat{j}+2z\hat{k}$

$|\nabla\phi| = \sqrt{4x^2+4y^2+4z^2}$

$= 2\sqrt{x^2+y^2+z^2}$ $\quad (\because x^2+y^2+z^2 = a^2)$

$= 2a$

$\therefore \hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2x\hat{i}+2y\hat{j}+2z\hat{k}}{2a}$

$= \frac{x\hat{i}+y\hat{j}+z\hat{k}}{a}$

$\text{curl}\vec{F}\cdot\hat{n} = (z\hat{j}-y\hat{k})\cdot\left(\frac{x\hat{i}+y\hat{j}+z\hat{k}}{a}\right)$

$= \frac{1}{a}[yz-yz]$

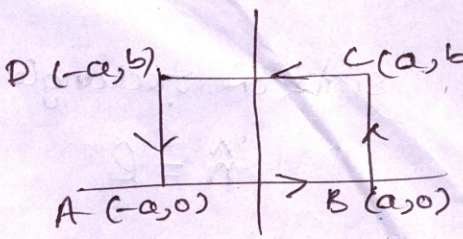
$= 0 \rightarrow (3)$

Thus $\int_C \vec{F}\cdot d\vec{r} = \iint_S \text{curl}\vec{F}\cdot\hat{n}\, ds$

Hence Stoke's theorem is verified.

(2) Verify Stoke's theorem for the vector $\vec{F} = (x^2+y^2)\hat{i} - 2xy\hat{j}$ (16)

taken round the rectangle bounded by the lines $x = \pm a$, $y = 0$ and $y = b$



Soln. By Stoke's theorem we have

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S \text{curl}\,\vec{F}\cdot\hat{n}\,ds$$

(i) Along $\oint_C \vec{F}\cdot d\vec{r} = \int_{C_1}\vec{F}\cdot d\vec{r} + \int_{C_2}\vec{F}\cdot d\vec{r} + \int_{C_3}\vec{F}\cdot d\vec{r} + \int_{C_4}\vec{F}\cdot d\vec{r}$

Along AB, $y = 0$, x varies from $-a$ to a

$$\therefore \int_{C_1}\vec{F}\cdot d\vec{r} = \int_{C_1}(x^2+y^2)\,dx - 2xy\,dy$$

$$= \int_{-a}^{a} x^2\,dx = \frac{x^3}{3}\Big]_{-a}^{a} = \frac{2a^3}{3}$$

Along BC $x = a$, $dx = 0$, y varies from 0 to b

$$\therefore \int_{C_2}\vec{F}\cdot d\vec{r} = \int_{0}^{b} -2ay\,dy = -ay^2\Big]_0^b = -ab^2$$

Along CD, $y = b$, $dy = 0$, $x \to a$ to $-a$

$$\therefore \int_{C_3}\vec{F}\cdot d\vec{r} = \int_{a}^{-a}(x^2+b^2)\,dx = \frac{-2a^3}{3} - 2ab^2$$

Along DA $x = -a$, $dx = 0$, $y \Rightarrow b$ to 0

$$\therefore \int_{C_4}\vec{F}\cdot d\vec{r} = \int_{b}^{0} 2ay\,dy = -ab^2$$

$$\therefore \oint_C \vec{F}\cdot d\vec{r} = \frac{2a^3}{3} - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 = -4ab^2 \to (1)$$

Now $\text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2+y^2) & -2xy & 0 \end{vmatrix} = -4y\hat{k}$

Since the rectangle lies in the xy plane, we have

$\hat{n} = \hat{k}$ and $ds = dx\,dy$

$\therefore \iint_S \text{curl}\vec{F} \cdot \hat{n}\, ds = \iint_S -4y\hat{k} \cdot \hat{k}\, dx\,dy$

$= -\int_{x=-a}^{a} \int_{y=0}^{b} 4y\, dy\, dx$

$= -\int_0^b 4y\, dy \times \int_{-a}^{a} dx$

$= -\frac{4y^2}{2}\Big]_0^b \times [a+a]$

$= -4ab^2 \quad \rightarrow (2)$

$\therefore$ From (1) and (2) Stoke's theorem is verified.

(3) Verify Stoke's theorem for $\vec{F} = (x^2-y^2)\hat{i} + 2xy\hat{j}$ in the rectangular region in the xy plane, the rectangle being given by (0,0), (a,0), (0,b) and (a,b)

Verify Stoke's theorem for $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ where S is the upper half of the sphere $x^2+y^2+z^2 = 1$ and C is the boundary.

Ans: $-\pi$ //

In xy plane, $x^2+y^2=1$, $z=0$

$x = \cos\theta$, $y = \sin\theta$

④ Verify Stoke's theorem for $\vec{F} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ where S is the upper half of the surface of the Sphere $x^2+y^2+z^2=1$ and C is the boundary.

Soln.: The enclosed curve C is for the hemisphere is the base circle whose equation is $x^2+y^2=1$

Parametric Eqn of circle

$\therefore$ we have $x = \cos\theta$, $y = \sin\theta$, $z = 0$

$dx = -\sin\theta\, d\theta$, $dy = \cos\theta\, d\theta$, $dz = 0$

Thus $\oint_C \vec{F} \cdot d\vec{r} = \oint_C ((2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$

$$= \int_{\theta=0}^{2\pi} (2\cos\theta - \sin\theta)(-\sin\theta\, d\theta)$$

$$= \int_{\theta=0}^{2\pi} (\sin^2\theta - 2\sin\theta\cos\theta)\, d\theta$$

$$= \int_{0}^{2\pi} (\sin^2\theta - \sin 2\theta)\, d\theta$$

$$= \int_{0}^{2\pi} \left(\frac{1-\cos 2\theta}{2} - \sin 2\theta\right) d\theta$$

$$= \frac{1}{2}\int_{0}^{2\pi} (1 - \cos 2\theta - 2\sin 2\theta)\, d\theta$$

$$= \frac{1}{2}\left[\theta - \frac{\sin 2\theta}{2} + 2\,\frac{\cos 2\theta}{2}\right]_0^{2\pi}$$

$$= \frac{1}{2}[2\pi - 0 + 1 - (0 - 0 + 1)] = \pi$$

$$\text{Curl}\,\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix}$$

$$= \hat{k}$$

OR
The projection of the sphere $x^2+y^2+z^2=1$ on xy plane is $x^2+y^2=1$, $z=0$. Thus unit normal vector to the surface is $\hat{k}$ ($\because x^2+y^2+z^2=1$) i.e $\hat{n}=\hat{k}$

$$\phi = x^2+y^2+z^2-1$$

$$\nabla\phi = 2x\hat{i}+2y\hat{j}+2z\hat{k}$$

$$|\nabla\phi| = 2\sqrt{x^2+y^2+z^2}$$

$$= 2$$

$$\therefore \hat{n} = x\hat{i}+y\hat{j}+z\hat{k} \quad \Rightarrow \quad \hat{n}\cdot\hat{k} = z$$

$$\text{Curl}\,\vec{F}\cdot\hat{n} = \hat{k}\cdot(x\hat{i}+y\hat{j}+z\hat{k})$$

$$= z$$

$$\iint_S \text{curl}\,\vec{F}\cdot\hat{n}\,ds = \iint_R \text{curl}\,\vec{F}\cdot\hat{n}\,\frac{dxdy}{|\hat{n}\cdot\hat{k}|}$$

$$= \iint_R z\,\frac{dxdy}{z}$$

$$= \iint_R dxdy$$

= Area of the circle $(x^2+y^2=1)$

$$= \pi(1)^2$$

$$= \pi$$

Hence Stoke's theorem is verified.

(5) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = 2y\hat{i} + 3x\hat{j} - z^2\hat{k}$ using Stoke's theorem, where C is the boundary of upper half of the surface of the sphere $x^2+y^2+z^2 = 9$.

Soln:- By Stoke's theorem, we have $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}\vec{F} \cdot \hat{n}\, ds$ $\rightarrow$ (1)

$$\vec{F} = 2y\hat{i} + 3x\hat{j} - z^2\hat{k}$$

$$\text{Curl}\,\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & 3x & -z^2 \end{vmatrix}$$

$$= 0\hat{i} - 0\hat{j} + (3-2)\hat{k}$$

$$= \hat{k}$$

The projection of the sphere $x^2+y^2+z^2=9$ on the xy plane is $x^2+y^2=9$. Thus the unit normal vector to the surface is $\hat{k}$. Thus $\hat{n} = \hat{k}$.

Now $\text{Curl}\,\vec{F} \cdot \hat{n} = \hat{k} \cdot \hat{k}$

$= 1$

$\therefore$ (1) $\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \iint_S 1\, ds$ $\quad \left[\because \iint_R dx\,dy = \text{Area}\right]$

$=$ Area of the circle

$= \pi(3)^2$

$= 9\pi$

(6) Evaluate by Stoke's theorem $\oint_C \sin z\,dx - \cos x\,dy + \sin y\,dz$

where C is the boundary of the rectangle $0 \le x \le \pi$,

$0 \le y \le 1$, $z = 3$

Soln:- By Stoke's theorem, we have $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n}\, ds$

Consider $\text{Curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin z & -\cos x & \sin y \end{vmatrix}$

$= \cos y\,\hat{i} + \cos z\,\hat{j} + \sin x\,\hat{k}$

Taking the projection in xy plane gives $\hat{n} = \hat{k}$

$\therefore \oint_C \vec{F} \cdot d\vec{r} = \iint_R (\cos y\,\hat{i} + \cos z\,\hat{j} + \sin x\,\hat{k}) \cdot \hat{k}\, \frac{dx\,dy}{|\hat{k} \cdot \hat{k}|}$

$= \iint_R \sin x\,dx\,dy$

$= \int_{x=0}^{x=\pi} \int_{y=0}^{y=1} \sin x\,dy\,dx$

$= \int_{x=0}^{x=\pi} \sin x\,dx \times \int_{y=0}^{y=1} dx$

$= -\cos x \Big|_0^\pi \times x \Big|_0^1$

$= -[-1-1] \times 1$

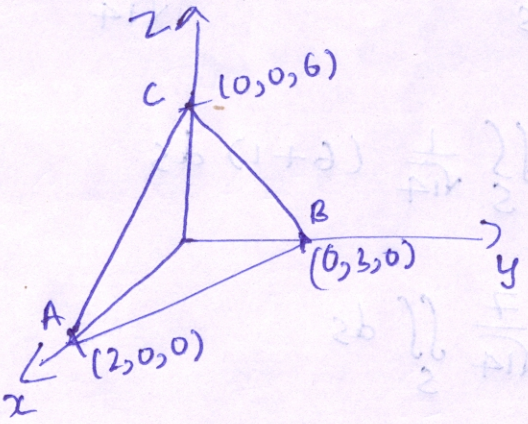
$= 2$ //

(7) Using Stoke's theorem evaluate $\int_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz$

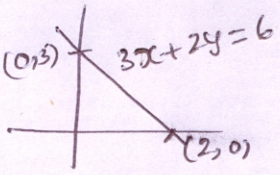
where C is the boundary of the triangle with vertices

$(2,0,0)$, $(0,3,0)$ and $(0,0,6)$.

Soln:-



Projection



Here $\vec{F} = (x+y)\hat{i} + (2x-z)\hat{j} + (y+z)\hat{k}$

$$\text{Curl}\,\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-z & y+z \end{vmatrix}$$

$$= 2\hat{i} + \hat{k}$$

Eqn of the plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$

$$\Rightarrow 3x + 2y + z = 6$$

$$\phi = 3x + 2y + z - 6$$

$$\nabla\phi = 3\hat{i} + 2\hat{j} + \hat{k}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{3\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{14}}$$

By Stoke's theorem, $\oint_C \vec{F}\cdot d\vec{r} = \iint_S \vec{F}\cdot\hat{n}\,ds$

Hence $\oint_C \vec{F} \cdot d\vec{r} = \int_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz$

$$= \iint_S \text{curl}\,\vec{F} \cdot \hat{n}\, ds$$

$$= \iint_S (2\hat{i}+\hat{k}) \cdot \frac{3\hat{i}+2\hat{j}+\hat{k}}{\sqrt{14}}\, ds$$

$$= \iint_S \frac{1}{\sqrt{14}} (6+1)\, ds$$

$$= \frac{7}{\sqrt{14}} \iint_S ds$$

$$= \frac{7}{\sqrt{14}} (\text{Area of the } \Delta ABC)$$

It is easy to take the projection of the plane in xy plane

$$\therefore ds = \frac{dx\,dy}{|\hat{n} \cdot \hat{k}|}$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \frac{7}{\sqrt{14}} \iint_R \frac{dx\,dy}{|\hat{n} \cdot \hat{k}|}$$

$$= \frac{7}{\sqrt{14}} \int_{x=0}^{x=2} \int_{y=0}^{y=\frac{6-3x}{2}} \frac{dy\,dx}{\frac{1}{\sqrt{14}}}$$

$$= 7 \int_0^2 \frac{6-3x}{2}\, dx$$

$$= \frac{7}{2} \left[6x - \frac{3x^2}{2}\right]_0^2$$

$$= \frac{7}{2}[12-6] = \frac{7}{2}[6] = 7 \times 3$$

$$= 21 \text{ //}$$